

# Knowledge Maps for Building Conceptual and Transferable CSS Knowledge

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## ABSTRACT

Despite the abundance of online resources for learning CSS, novice web developers struggle to develop the expert intuition of choosing the best CSS technique to build a given layout. We present Knowledge Maps (KM), a tool that allows users to build transferable and conceptual knowledge of CSS techniques. By interactively exploring professional websites and by categorizing visual features of those sites and the relevant CSS techniques used to create them, KM users discover the relevant similarities, differences, and use cases of various CSS techniques in the process, developing the knowledge that characterizes experts. In a study where 6 users learned from conventional CSS tutorials and 7 users learned through KM, KM users identified the appropriate CSS to build a set of layouts with a 48% increase in accuracy as compared to a 15% increase for non-KM users and showed development of transferable knowledge through KM.

## KEYWORDS

CSS; Comparison and Contrast; Web Development; Knowledge Transfer; Contextual and Conceptual Understanding

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## 1 INTRODUCTION

As the world is increasingly run by software, there is a higher demand for professional software developers. While there exists a growing number of novice web developers, there is a shortage of experts. This is because existing resources, such as online tutorials (Codecademy, Stack Overflow, and W3Schools), coding boot camps, and computer science degrees do not necessarily bridge the gap between novice and expert developers.

Novices understand explicit CSS syntax and can use specific CSS techniques in the context in which they were taught. Experts, along

with understanding CSS syntax, have deep conceptual knowledge of the various CSS techniques and their use cases. As a result of their years of experience, experts develop an intuition for understanding which CSS techniques to use when and which techniques are appropriate to build out the layout of any given website. Novices are not able to build these complex connections nor develop this intuition from completing tutorials which only teach CSS given in one context. Aside from tutorials, Chrome Developer Tools can be used to understand the CSS techniques that are responsible for any given professional website's layout. However, Chrome Developer Tools are much too complex for beginners to utilize and gain knowledge from effectively [4].

There have been contributions in this space from a previous iteration of this project. In the previous iteration, it was shown that (1) users were able to better understand how specific CSS properties affect the visual features of an example by tinkering with the interactive CSS properties; (2) code highlighting is effective in helping learners identify and form a basis for analyzing visual and code differences in two examples [7]. In this previous iteration, users were able to formulate a rationale for why the author in each example chose their professional CSS technique. However, they did not translate this knowledge to a broader mental model for how each CSS technique can be used in general.

We present a new iteration of Knowledge Maps (KM), a platform to help learners understand the tradeoffs between different CSS techniques and the various use cases associated with those techniques. The conceptual contribution of this paper is that exploring various website examples and then grouping the visual attributes of those websites with the CSS techniques used to produce them creates a more generalized model of CSS knowledge that allows for conceptual understanding of the use cases and applications of each CSS technique.

The mechanisms utilized on KM include, as before, highlighted code properties and interactive CSS code in a new interface with expanded features, along with the addition of a format for creating a general knowledge map using matching and category creation. As stated in *How Learning Works* [1] and research from the National Research Council [2], how students organize knowledge influences how they learn and apply what they know, highlighting the importance of the easily digestible knowledge map.

During a user test, KM helped users create a general knowledge map for CSS grid and apply that map to specific website examples to identify the CSS grid techniques used to build each site with three times as much accuracy improvement as compared to users

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who used alternative methods of learning CSS grid. KM's mechanisms were useful in helping learners identify key differentiating features between examples and turn those different features into categories for different visual outcomes. Users could then identify which CSS grid techniques were responsible for producing each visual outcome.

## 2 RELATED WORK

Users do not care specifically about learning CSS grid in general as much as they care about being able to build complex layouts. Our central challenge is to teach CSS grid in a way that the knowledge gained through using KM transfers to the act of building complex web layouts without the help of the tool. This leads to the central problem faced by most research in the learning sciences: the problem of transfer. Psychologist Robert Haskell, after conducting a vast review of the literature on transfer in learning, says, "Unfortunately, transfer is also something that, despite more than a century of intense work and research, has largely failed to occur in formal education" [3]. Haskell suggests that transfer tends to be more difficult when our knowledge is limited, and that as we delve deeper and gain more experience in a subject area, we are more likely able to apply our knowledge outside the narrow contexts in which it was originally learned. Applying this idea to Knowledge Maps, we propose a solution to overcome the problem of transfer through "directedness": directly modifying CSS syntax and seeing the results of those modifications rather than passively reading a tutorial, giving users a deeper dive into how CSS grid techniques work.

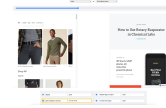
Many current solutions exist to help novice developers understand CSS techniques. Tutorons helps learners build contextual understanding by providing learners with in-line explanations of CSS properties that they may not be familiar with. Ply identifies the relevance of specific CSS properties for a selected visual feature so users can isolate the relevant code from all of the source code. Both solutions help learners understand more about particular CSS techniques, however, neither are sufficient to help learners compose their learnings to build generalized complex CSS layouts.

Lastly, KM instructs users to build a knowledge map, which is a bipartite graph mapping visual outcomes to the corresponding CSS techniques used to produce them. This is inspired by research from Schwartz and Bransford [6] arguing that by using contrasting cases while learning, learners can identify specific features differentiating each case to generate knowledge structures for conceptual understanding. Additionally, Marcia C Linn and Michael J Clancy [5] found that articulating and reflecting on what one has learned is critical to learning from examples. Website example pairings that show the same CSS technique used in two contrasting layouts or the same layout produced by two different CSS techniques allow users to identify and create visual categories of those layouts in the knowledge map and then match the corresponding CSS techniques used to create them. The activity of building a knowledge map introduces other benefits. As explained in How Learning Works [1], how students organize knowledge influences how they learn and apply that knowledge. The knowledge map learners build provides scaffolding to learn about CSS layouts in the way an expert would organize their knowledge about CSS layouts.

## 3 SYSTEM DESCRIPTION

KM consists of two separate phases: the explore phase and the knowledge map phase.

### 3.1 The Explore Phase



**Figure 1: The split screen design to compare two examples at once**

To explore the sites, users are able to resize the website windows using the sliding bar. These websites were chosen because they had similar layouts implemented by different CSS techniques or different layouts implemented by the same CSS techniques. Five websites were chosen to ensure enough variety and effective comparison.



**Figure 2: The code editor in the explore phase**

The code editor (Fig. 2) has a block for each CSS declaration. Declaration blocks are highlighted in yellow if the same CSS property is used in both examples but the value of that property is different; blocks are highlighted in green if the same CSS property and the same value of that property are used in both examples. There are affordances for adding new CSS declarations to the parent element as well as editing existing declarations. When edits are made to the editor by the user, the structured data is synthesized into valid CSS, and using the browser postMessage API, the example's output is updated. Javascript is added to the example website in order to receive modified CSS by the user and then apply that modified CSS to the website.

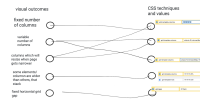
### 3.2 The Knowledge Map Phase

Users are required to build out a knowledge map in the knowledge map phase. From exploring the sites, users create categories of visual outcomes they notice. Examples of visual outcomes are: (1) the number of columns changes based on the window size of the site; or (2) the number of columns stays the same regardless of window size. Users then write the relevant CSS techniques used to produce these visual outcomes and match them together by drawing a line connecting one or more visual outcome circles on the left to one or more CSS technique circles on the right. Users go back and forth between exploring the sites and building out their knowledge map.

## 4 USER STUDY AND EVALUATION

Our user study seeks to answer the following questions:

- (1) Do users have a better understanding of the tradeoffs and use cases of CSS grid after using the KM tool as compared to alternative ways of learning CSS?



**Figure 3: A completed knowledge map by one of our participants**

- (2) Does building out the knowledge map in the KM tool create conceptual understanding that allows users to apply their knowledge onto new contexts?

#### 4.1 Setup

We recruited 13 computer science students from Northwestern University who had a basic knowledge of CSS. These students had all taken an introductory HTML/CSS/Javascript course, although some had more experience with CSS than others. We held a 32-minute study that consisted of a pre-test, a learning phase, and a post-test. The participants were randomly divided into two groups. Group 1 used the KM tool during the learning phase, while Group 2 used traditional methods of learning CSS grid.

Each participant in both groups completed the same survey testing their previous knowledge of CSS grid. They were asked to examine three professional websites not included in the KM tool that used CSS grid as the main element of their layout and describe the visual layout and the CSS techniques they believed to be producing the layout. We scored the participants' accuracy on the pre-test by comparing each participant's answers against our rubric. Our rubric outlined the CSS techniques responsible for each website's layout. Each contributing CSS technique or value was given a point value (either 1 or 2). The participant received the corresponding number of points for each CSS technique or value they correctly identified. The participant's score for each site was calculated to be the number of points they achieved over the total number of points possible for that site.

In the learning phase of the study (16 minutes), Group 1 used the KM tool while Group 2 acted as a control group, utilizing traditional methods of learning CSS grid. Participants in Group 1 utilized the explore phase and knowledge map phase of KM described above. Participants in Group 2 used traditional methods of learning CSS grid in the allotted time, such as W3 Schools, Stack Overflow, exploring grid websites, and inspecting them with Chrome Developer Tools. They were told to focus specifically on CSS grid techniques. They did not build out a knowledge map.

In the post-test (8 minutes), participants in Group 1 and Group 2 were presented with the same survey as in the pre-test. They were shown the same three websites and asked to identify the CSS techniques responsible for producing each website's layout. Both groups were allowed to use their resources when completing the post-test: the knowledge map for Group 1 and online sources for Group 2. Post-test results were scored using the same rubric as the pre-test. Each participant ended up with 6 scores: 3 scores for the 3 pre-test sites and 3 scores for the same 3 post-test sites. To receive more data on the participants' understanding, we asked for verbal feedback after the study as well. To analyze the test results, we

summed up each participants' scores for the 3 pre-tests and the 3 post-tests.

## 5 STUDY FINDINGS

### 5.1 Results for RQ1

**By exploring and tinkering with examples side by side and building out a knowledge map using KM, participants gained a better understanding of the tradeoffs and use cases of CSS grid than when using alternative methods of learning CSS.** The average increase in score from pre-test to post-test in the control group was 15%, whereas the average increase in score for the group that used KM was 48%. This validates our hypothesis that KM is more effective at teaching the use cases of CSS techniques, specifically CSS grid, than conventional methods of learning CSS. In a post-study interview, one participant shared about KM that he found it "really interesting how [he] could learn about many different CSS attributes at once. [He] especially liked being able to play around with the CSS values and see what visual change resulted from it." Another participant shared that through KM, he "found a really clear workflow of finding the unique CSS in the tool and copying what that usage was into the [knowledge map], and reflecting on how that connected to the unique visual characteristics [he] was seeing."

### 5.2 Results for RQ2

**By guiding participants to build out a knowledge map, participants were able to form transferable knowledge about which visual outcomes could be produced by which CSS techniques, going beyond making conclusions about a particular example and thus forming conceptual knowledge.** The group that used KM had a 48% increase in score when asked to identify the relevant CSS code producing the layouts of a group of websites. The group of websites that the participants were tested on was different from the group websites they studied in the KM tool, proving that the knowledge they gained from the tool was able to be transferred to a completely different set of examples and is thus transferable. One participant who scored particularly high in the post-test shared that "when [he] started connecting [visual outcomes] to multiple properties, [he] realized that there were different ways of controlling a property, like grid spacing... this helped [him] see patterns and multiple points of similarity across examples." We noticed that students who had more in-depth knowledge maps, like his, performed better on the post-test, pointing to the effectiveness of an in-depth knowledge map in building transferable and conceptual knowledge.

## 6 DISCUSSION AND FUTURE WORK

We noted that the lowest scoring users only mentioned that they would use CSS grid to achieve certain layouts without specifying which particular syntax they would use to achieve a certain outcome. Many learners did not grasp the additional layers of complexity within CSS grid techniques. For example, the grid specification introduces the repeat function, which takes possible values of: auto-fill, auto-fit, and others. The KM platform did not have processes to nudge users to explore these individual techniques more closely. The tool was best utilized by people who had adequate experience

with CSS but not grid specifically. People who were unable to extract value from the tool had very limited experience with CSS, which indicated a failure in recruiting appropriate users for the study.

## 6.1 Limitations

Although the study results suggest that learners can grasp the basics of CSS grid using KM, the analysis does not show that participants were able to learn particular CSS grid techniques deeply. For example, no one in the study fully grasped how the fractional units introduced by CSS grid work, which was a relevant contributor to the layout of one of the websites in the tool.

The study had a limited number of users. It is possible that the KM tool's specific interfaces may fit certain types of learners, such as visual learners, more than others. Because there were a limited number of users, it is possible that results were more positively skewed towards learners who the tool's style favored.

Additionally, the example websites in the KM tool used similar CSS grid techniques to the ones in the pre and post-tests. In this way, the tool guided users to focus on these specific techniques. Users in Group 2, the control group, had less guidance on which CSS grid techniques to focus on and focused on CSS grid as a whole.

## 6.2 Future Work

One limitation of KM is that in its current iteration, there is no integration between the explore phase and the knowledge map phase. Users extract value from the explore phase without any support or guidance from the knowledge map phase. Future work would integrate the two phases to provide feedback to users as they completed their knowledge map.

We hope to expand KM to include more websites to learn more techniques than just CSS grid. By adding another CSS technique, such as flex, users can explore similar layouts that are produced by different CSS techniques (grid vs. flex) to gain a better understanding of the tradeoffs and use cases of each technique. We would like to add more CSS techniques to allow users to gain full knowledge of CSS, eventually enabling them to both be able to identify appropriate CSS techniques used to build professional websites and to identify the appropriate CSS techniques to use when building their own web development projects.

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## REFERENCES

- [1] Susan A. Ambrose. 2010. *How Learning Works : Seven Research-Based Principles for Smart Teaching*. Wiley, Hoboken.
- [2] National Research Council. 2001. *Educating Children with Autism*. The National Academies Press, Washington, DC. <https://doi.org/10.17226/10017>
- [3] Robert E. Haskell. 2001. *Transfer of learning : cognition, instruction, and reasoning*. Academic Press, San Diego, Calif.
- [4] Sarah Lim. 2017. Ply: Visual Regression Pruning for Web Design Source Inspection. *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (2017).
- [5] Marcia C. Linn and Michael J. Clancy. 1992. The Case for Case Studies of Programming Problems. *Commun. ACM* 35, 3 (mar 1992), 121–132. <https://doi.org/10.1145/131295.131301>
- [6] Daniel L. Schwartz and John D. Bransford. 1998. A Time For Telling. *Cognition and Instruction* 16, 4 (1998), 475–5223.
- [7] Daniel Zhu and Salome Wairimu Kariuki. 2020. Knowledge Maps: Building Conceptual CSS Knowledge Through Comparison (*CHI EA '20*). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3334480.3381444>